

Quantum Pointer-State Selection

A MAAT v1.4 Defect-Field Benchmark in a Non-Commuting Lindblad System

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Abstract

Paper 50c reformulated MAAT structural selection as a multi-scale defect-field framework rather than a single global scalar compression. The present paper tests the same v1.4 idea in a minimal quantum setting: pointer-state robustness in a non-commuting Lindblad system.

We generate a fixed-seed ensemble of 3500 qubit states and open-system generators containing three competing channels: z -dephasing, x -dephasing, and amplitude damping. The target is the survival fidelity of the initial pure state after Lindblad evolution. All features are computed from the initial state and generator parameters, not from the evolved final state, so the benchmark is leakage-free.

Scalar MAAT supports retain moderate predictive power,

$$R_{\text{scalar}}^2 = 0.5383 \pm 0.0151,$$

but the v1.4 field-feature extension substantially improves prediction:

$$R_{\text{v1.4}}^2 = 0.7424 \pm 0.0057.$$

Simple baselines are much weaker, with

$$R_{\text{rates}}^2 = 0.0542, \quad R_{\text{domain}}^2 = 0.1362.$$

The gain over scalar MAAT is

$$\Delta R^2 = 0.2041.$$

The result is not a derivation of quantum measurement or a full decoherence theory. It is a toy-model benchmark showing that pointer robustness is not captured by decoherence rates alone, but by a structured interaction between basis alignment, channel conflict, damping pressure, coherence exposure, and robustness closure.

1 Purpose and Status

Paper 49 identified quantum pointer-state selection as a missing universality test for the MAAT programme. Papers 50–50c then showed, in SAT, that scalar compression can destroy important local defect-field information. The present paper asks whether the same distinction appears in a minimal quantum benchmark:

Do v1.4 field features improve pointer-robustness prediction beyond scalar MAAT compression and simple quantum-domain baselines?

The scope is deliberately modest. This is not a theory of wavefunction collapse, not a derivation of the Born rule, not an experimental fit, and not a full environment-induced superselection treatment. It is a leakage-free toy benchmark in which structural features are compared against explicit baselines.

2 Open-System Benchmark

The system is a single qubit with initial pure state

$$|\psi(\theta, \phi)\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle. \quad (1)$$

The Hamiltonian is

$$H_{\text{sys}} = h_z \sigma_z + h_x \sigma_x. \quad (2)$$

The Lindblad channels are

$$L_z = \sqrt{\gamma_z} \sigma_z, \quad (3)$$

$$L_x = \sqrt{\gamma_x} \sigma_x, \quad (4)$$

$$L_m = \sqrt{\gamma_m} \sigma_-. \quad (5)$$

The density matrix evolves according to

$$\dot{\rho} = -i[H_{\text{sys}}, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right). \quad (6)$$

The target is the pointer-survival fidelity

$$y = \langle \psi | \rho(t) | \psi \rangle. \quad (7)$$

3 Feature Construction

The scalar MAAT supports are:

$$H = \frac{1}{1 + \|[H_{\text{sys}}, \rho]\|}, \quad (8)$$

$$B = \frac{1}{1 + S_{\text{pop}}(\rho)}, \quad (9)$$

$$S = \frac{1}{1 + w_{\text{offdiag}}(\rho)}, \quad (10)$$

$$V = \frac{\gamma_z |b_z| + \gamma_x |b_x| + \gamma_m \max(b_z, 0)}{\gamma_z + \gamma_x + \gamma_m + \epsilon}. \quad (11)$$

Here $\vec{b} = (b_x, b_y, b_z)$ is the Bloch vector. The response and robustness closures are

$$R_{\text{resp}} = (HBV)^{1/3}, \quad R_{\text{rob}} = \min(R_{\text{resp}}, (HBSV)^{1/4}). \quad (12)$$

The scalar score is

$$F_{\text{MAAT}} = -\log H - \log B - \log S - \log V - \log R_{\text{rob}}. \quad (13)$$

The v1.4 field-feature extension adds:

- channel conflict between non-commuting dephasing channels;
- damping pressure from the amplitude-damping channel;
- coherence exposure to dephasing and damping;
- x -exposure caused by x -dephasing acting against z -alignment.

These features are not final microscopic principles. They are operational defect-field diagnostics chosen to test whether field-level channel structure improves prediction beyond scalar compression.

4 Baselines

Six models are compared by five-fold cross-validated random forests:

Model	Features
MAAT scalar/supports	$H, B, S, V, R_{\text{resp}}, R_{\text{rob}}, F_{\text{MAAT}}$
MAAT v1.4 field features	scalar supports plus channel-conflict and exposure features
Rates only	$\gamma_z, \gamma_x, \gamma_m, t$
Basis geometry only	$ b_z , b_x , b_y $, off-diagonal weight
Hamiltonian only	H, h_x, h_z
Domain combo	rates, basis geometry, and Hamiltonian parameters

5 Results

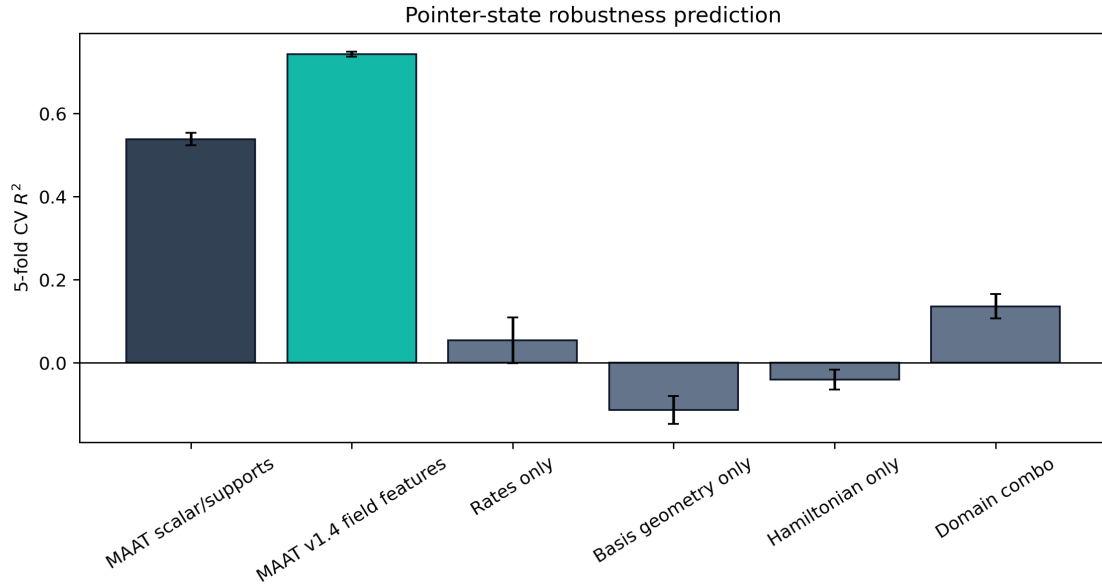


Figure 1: Cross-validated pointer-robustness prediction. The v1.4 field-feature model substantially improves over scalar MAAT and over simple domain baselines.

Table 1: Five-fold cross-validated random-forest prediction of pointer survival fidelity.

Model	R^2	Std.
MAAT scalar/supports	0.5383	0.0151
MAAT v1.4 field features	0.7424	0.0057
Rates only	0.0542	0.0547
Basis geometry only	-0.1134	0.0331
Hamiltonian only	-0.0406	0.0241
Domain combo	0.1362	0.0291

The key improvements are:

$$\Delta R_{v1.4\text{--scalar}}^2 = 0.7424 - 0.5383 = 0.2041, \quad (14)$$

$$\Delta R_{v1.4\text{--domain}}^2 = 0.7424 - 0.1362 = 0.6062. \quad (15)$$

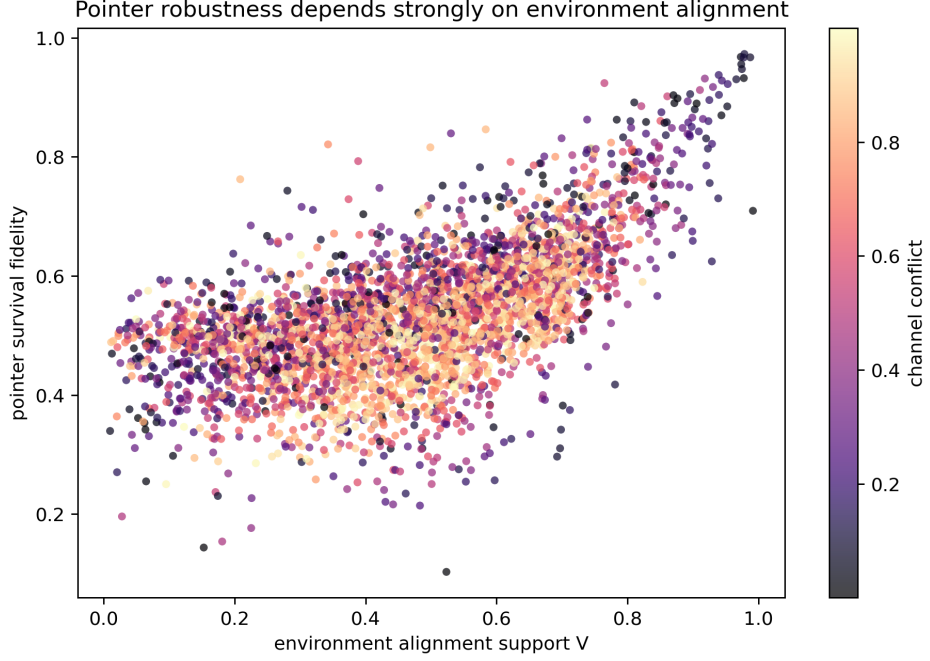


Figure 2: Environment alignment support V versus pointer survival fidelity. The color encodes channel conflict.

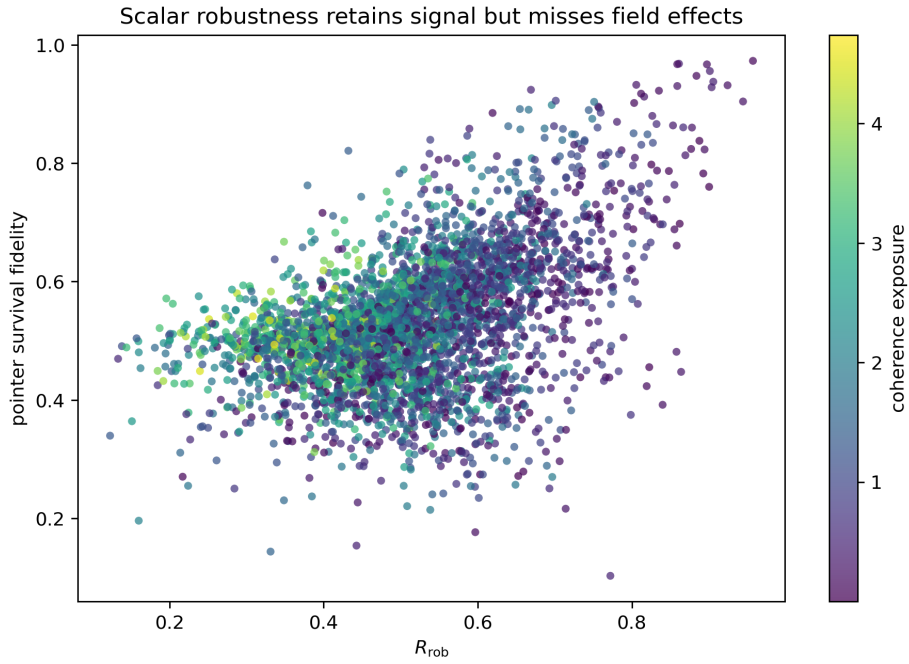


Figure 3: Robustness closure R_{rob} versus pointer survival fidelity. Scalar robustness retains signal but misses field-level exposure structure.

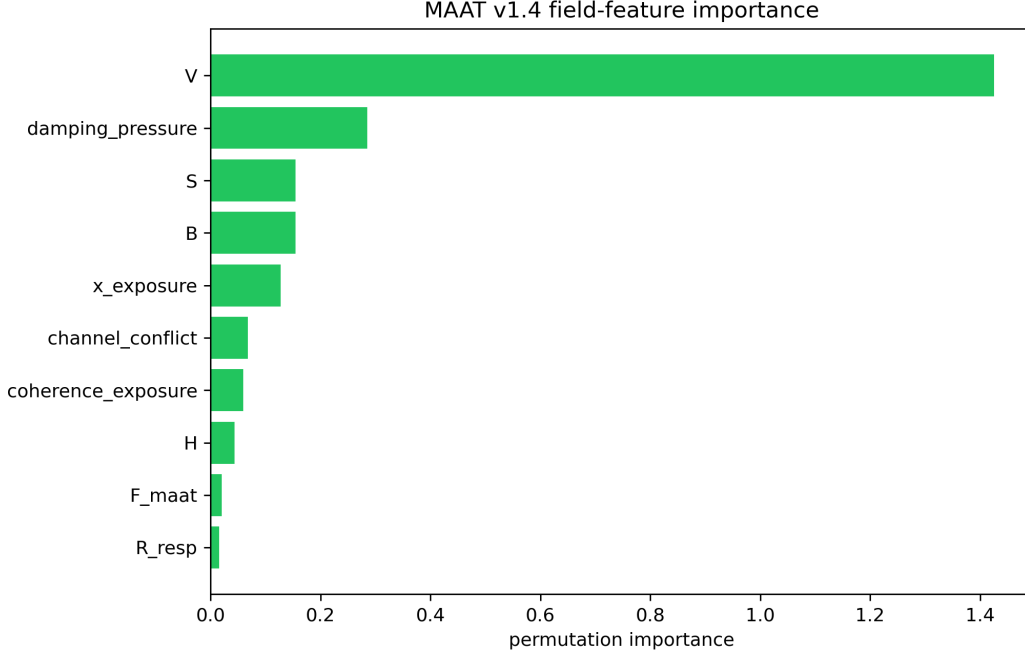


Figure 4: Permutation feature importance for the MAAT v1.4 field-feature model.

The most important v1.4 features are:

Feature	Permutation importance
V	1.4252
Damping pressure	0.2855
S	0.1550
B	0.1544
x -exposure	0.1273
Channel conflict	0.0684
Coherence exposure	0.0597
H	0.0438

6 Interpretation

The benchmark gives a positive but bounded result. Scalar MAAT already retains nontrivial pointer-state information. This is visible both in the cross-validated score and in scalar correlations:

$$\rho_S(V, y) = 0.6034, \quad \rho_S(R_{\text{rob}}, y) = 0.4306, \quad \rho_S(F_{\text{MAAT}}, y) = -0.3771.$$

However, the v1.4 field extension adds information not captured by the scalar supports alone.

Physically, this is plausible. Pointer robustness in a non-commuting open system is not determined by rates alone. It depends on how the state sits relative to multiple incompatible channels, how strongly coherence is exposed, and whether damping pressure aligns or competes with the candidate pointer basis.

The key lesson is:

Pointer-state robustness is a field-structured compatibility problem, not a rate-only problem and not merely a scalar robustness problem.

7 Limitations

Several limitations remain:

- the system is a single-qubit toy model;
- the environment has only three hand-selected Lindblad channels;
- the v1.4 features are operational diagnostics, not derived microscopic postulates;
- no multi-qubit decoherence, entanglement, or measurement apparatus is included;
- no experimental dataset is used;
- stronger quantum-information baselines should be added in future work.

8 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/quantum_pointer_state_selection_paper51

Run:

```
python3 quantum_pointer_state_selection_paper51.py
```

The script writes:

- `outputs_paper51/paper51_pointer_instances.csv`;
- `outputs_paper51/paper51_model_comparison.csv`;
- `outputs_paper51/paper51_scalar_correlations.csv`;
- `outputs_paper51/paper51_v14_feature_importance.csv`;
- `outputs_paper51/paper51_summary.json`;
- `outputs_paper51/fig1_model_comparison.png`;
- `outputs_paper51/fig2_V_support_vs_fidelity.png`;
- `outputs_paper51/fig3_Rrob_vs_fidelity.png`;
- `outputs_paper51/fig4_v14_feature_importance.png`.

The benchmark generates synthetic quantum toy-model data with a fixed random seed. No external quantum dataset is redistributed.

9 Conclusion

Paper 51 provides the first quantum-domain test of the MAAT v1.4 defect-field update. In a non-commuting Lindblad benchmark, scalar MAAT supports retain moderate pointer-robustness signal, but v1.4 field features substantially improve prediction and strongly outperform simple rate-only and domain-combo baselines.

The result is not a measurement theory. It is a reproducible toy benchmark showing that structural selection can be tested in open quantum systems and that field-level compatibility features matter.

References

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